Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 32)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 1 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

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0 1 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops .....

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Roundup Ready soya .....

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Golden rice .....

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0 1 . 3 Explain **four** concerns about GM crops.

[4 marks]

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0 1 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

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0 2 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 2 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 2 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 3 A student investigated the effect of pH on the germination of cress seeds.

0 3 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

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0 3 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

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0 3 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 3 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

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0 4 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 4 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 4 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 5 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

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0 5 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

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0 5 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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0 6 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 6 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 6 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 6 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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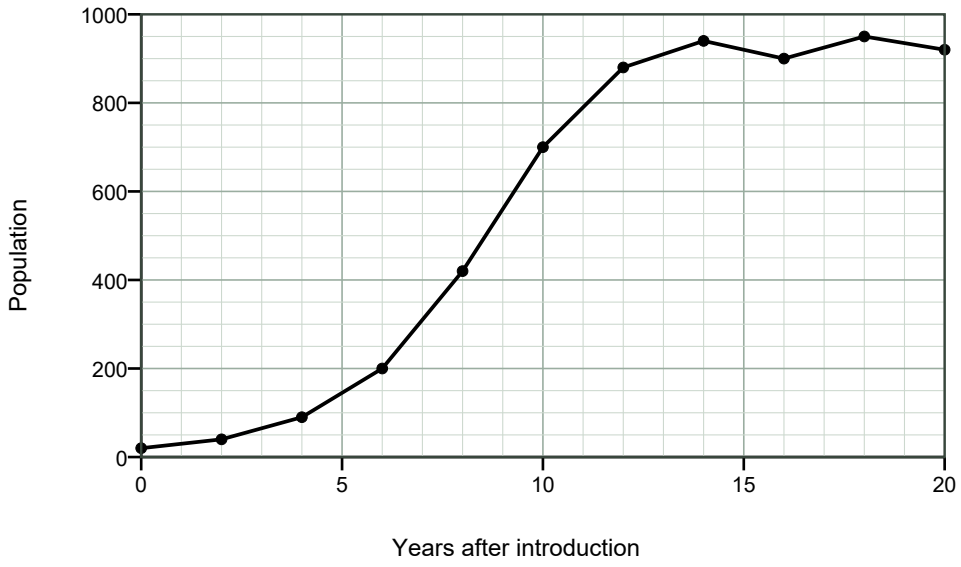
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0 7 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 7 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 7 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

0 7 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 7 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 8 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

0 8 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor

Introduced predator

Introduced pathogen

0 8 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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0 9 Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha^{-1}	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 9 . 1 Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

0 9 . 2 Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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0 9 . 3 Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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1 0 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

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1 0 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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1 0 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

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1 0 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

11 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

11 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS